

VERIFIED RESULTS · FOOD PROCESSING SITE

Patented capital equipment cut refrigeration energy **41.5%** at a food processing site.

With zero operational disruption and no impact to setpoints, product quality, or workflow.

41.5%

REFRIGERATION REDUCTION

11.7%

SITE-WIDE (MAIN INCOMER)

130,685

KWH ANNUALIZED

8units

12–73% RANGE

SOLUTION

Patented enPact Refrigeration & Cooling optimization. Deployed across multiple high-load refrigeration units to improve heat-exchange, reduce compressor run-time, and stabilize temperatures — without disruption to production, cold-store operation, or workflow.

- Improve compressor operating efficiency
- Enhance heat-exchange performance
- Stabilize temperatures across units
- Deliver measured, ON/OFF-validated savings
- Zero disruption to production or workflow

CHALLENGE

A food processing facility with cold-store, processing, and packing areas faced rising electricity costs. Refrigeration operated continuously and represented a significant share of site demand. Key constraints:

- Continuous refrigeration demand across multiple units
- High operational temperatures driving excessive compressor cycling
- Refrigeration imbalance between units causing inefficiencies
- Requirement for proven, validated savings — not estimated reductions
- No option for production downtime or operational disruption

Results.***Refrigeration + site-wide savings.***

Live measurement across 8 refrigeration units using ON/OFF comparison periods validated substantial reductions with zero operational disruption.

- ▶ 41.5% refrigeration reduction (site total, 8 units)
- ▶ 11.7% site-wide reduction at the main incoming supply
- ▶ 358 kWh/day saved · 130,685 kWh/yr annualized
- ▶ Individual unit range: 12–73% reduction (2 units neutral)
- ▶ Result exceeded the pre-trial quoted value of 36,202 kWh
- ▶ Zero operational disruption during trial or deployment

VERIFIED METRICS

Technology	enPact R&C
Refrigeration reduction	41.5%
Site-wide reduction	11.7%
Annualized savings	130,685 kWh
Unit range	12–73% (2 neutral)
Units instrumented	8 refrigeration units

Refrigeration is the driver. Because refrigeration was the highest-load system onsite, improving compressor efficiency delivered large downstream effects — lowering overall electrical demand well beyond the cold-store areas. The unit-level range (12–73%) reflects the natural variance in load and configuration; the site total was validated at 41.5%.

CONCLUSION

enPact Refrigeration & Cooling delivered a 41.5% refrigeration reduction and 11.7% site-wide reduction with zero operational disruption — substantially exceeding the pre-trial business case.

ESTATE ROLLOUT PROJECTION

Applied across the client's three sites: 392,000 kWh/yr savings, consistent 11–12% reductions per location — with no operational disruption.

Interested in achieving similar savings across your estate?Golden Grid Solutions · info@GoldenGSolutions.com · 951-744-5092 · goldengridsolutions.com